

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0960

Airline 230 kV
March 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0960. The Interconnection Customer proposes a 201.6 MW solar photovoltaic facility interconnecting at the Airline 230 kV substation in the Ohio area. The request was received on March 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 408 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,117,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STEVENS RD / CONESVILLE STA 230kV	MUSKINGUM RIV / CONESVILLE STA 500kV Line	109.9	624	6.2
CONESVILLE STA / SUNNYVALE 230kV	DRESDEN / CONESVILLE STA 500kV L	102.2	566	6.4
SUNNYVALE / DRESDEN 230kV line	DRESDEN / CONESVILLE STA 500kV	101.3	602	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STEVENS RD / CONESVILLE STA 230kV	Reconductor line	\$11,711,000	3% (\$394,000)
CONESVILLE STA / SUNNYVALE 230kV	Reconductor line	\$11,303,000	5% (\$525,000)
SUNNYVALE / DRESDEN 230kV	Reconductor line	\$4,265,000	5% (\$198,000)
	Total allocated to project		\$1,117,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.